

Statement of Overall Scientific Interests and Research Goals

Overall Scientific Interests.....

My overall research interests lie in the development and validation of methodologies for avian bioacoustic monitoring, with a particular focus on integrating passive acoustic monitoring (PAM) with machine learning and statistical approaches for ecological inference. Over the years, I have engaged in a wide range of bioacoustic work through international collaborations, including serving as a field recordist and collecting vocalizations from over 300 bird species across Taiwan, Canada, Lithuania, and Siberia. These recordings have been used in museums and art exhibitions, popular science articles, musical compositions, audiobooks, and podcasts. Through these experiences, I developed a deep appreciation for the communicative and cultural value of bird sounds, as well as for how listening can shape our understanding of nature. During this period, I began to integrate my background in physics (bachelor's degree) and biometrics (master's degree) with my long-standing interest in bird sounds. This convergence shaped my pursuit of avian bioacoustics as a quantitative and methodologically driven research field. In particular, I was drawn to developing reproducible methods that enable avian bioacoustic data to be used reliably for ecological and conservation applications.

This methodological orientation strongly shaped my PhD research, which I began in 2021 during a period of rapid expansion in passive acoustic monitoring. This expansion was driven by the increasing availability of low-cost recording hardware, such as AudioMoth, alongside open-source machine-learning tools for automated species identification from audio data, such as BirdNET. Despite their widespread adoption, these technologies remain relatively recent, and clear analytical guidelines and standardized workflows for generating reliable ecological datasets are still limited. To address this gap, my PhD thesis included a major chapter devoted to developing a detailed workflow for setting species-specific BirdNET confidence thresholds to obtain reliable detections. I then applied this workflow across multiple ecological scales—within-species, single-species, and multi-species—to demonstrate how PAM data, when paired with appropriate statistical modeling, can support robust and interpretable ecological inference. Collectively, this work sought to bridge the gap between automated sound detection and ecological interpretation, while providing generalizable guidance for the effective use of PAM in ecological research.

Beyond applying passive acoustic monitoring to ecological questions, I have a strong interest in developing open-source scientific tools, particularly R packages, as a means of supporting reproducibility and accessibility in bioacoustics research. Through my engagement with rOpenSci, an organization that fosters open and reproducible research using shared data and reusable software, I developed the R package `bbsTaiwan`, which enables users to access and manage Taiwan Breeding Bird Survey data within R. Building on this work, I developed `birdnetTools`, an R package that standardizes workflows for validating, processing, and analyzing BirdNET detections. I led the software design in close collaboration with the BirdNET team at Chemnitz University of Technology and the K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology. `birdnetTools` has received growing attention from the bioacoustics community and is currently supported by an R Consortium Infrastructure Steering Committee Grant to expand its functionality toward occupancy modeling and other population-level applications. This work reflects my broader interest in translating advances in bioacoustics into sustainable, widely usable research infrastructure.

Sound, particularly avian bioacoustics, has played a central role in my research trajectory, serving simultaneously as a medium of communication, a powerful approach for documenting biodiversity, and a rich data source for ecological analysis. Looking forward, I aim to continue expanding the use of passive acoustic monitoring for long-term biodiversity assessment, while developing robust tools and scalable statistical methods that enhance the accuracy, accessibility, and ecological relevance of bioacoustic data for ecology and conservation.

Research Goals.....

Overview: Great Cormorant (*Phalacrocorax carbo*) colonies are known for their substantial ecological impacts on terrestrial ecosystems, altering habitat structure, resource availability, and perceived predation risk, and thereby reshaping the diel and seasonal activity of co-occurring birds and mammals. Understanding these activity shifts is essential for evaluating the broader ecosystem effects of cormorants. This research will assess diel and seasonal activity across cormorant density gradients by combining passive acoustic monitoring of birds with camera trapping of mammals, providing cross-taxonomic insights to inform evidence-based management of cormorant colonies.

Introduction: Great Cormorants are avian top predators and are the most widely distributed of all cormorant species globally¹. Given their advanced fish-predation skills, they were largely exterminated in the past^{2,3}; however, a substantial population increase has been observed over the past 40 years, largely due to legal protection across countries and their high degree of ecological adaptation^{4,5}. This population increase has raised concerns about the environmental consequences of their presence, as these birds nest colonially in large numbers (from a few to several thousand individuals⁶), with up to fifty nests on a single tree⁷, potentially triggering drastic environmental changes. Changes in terrestrial ecosystems, particularly those caused by faecal deposition and carrion fall, include soil fertilization, shifts in plant species composition and biomass, and increased scavenger and predator activity⁷.

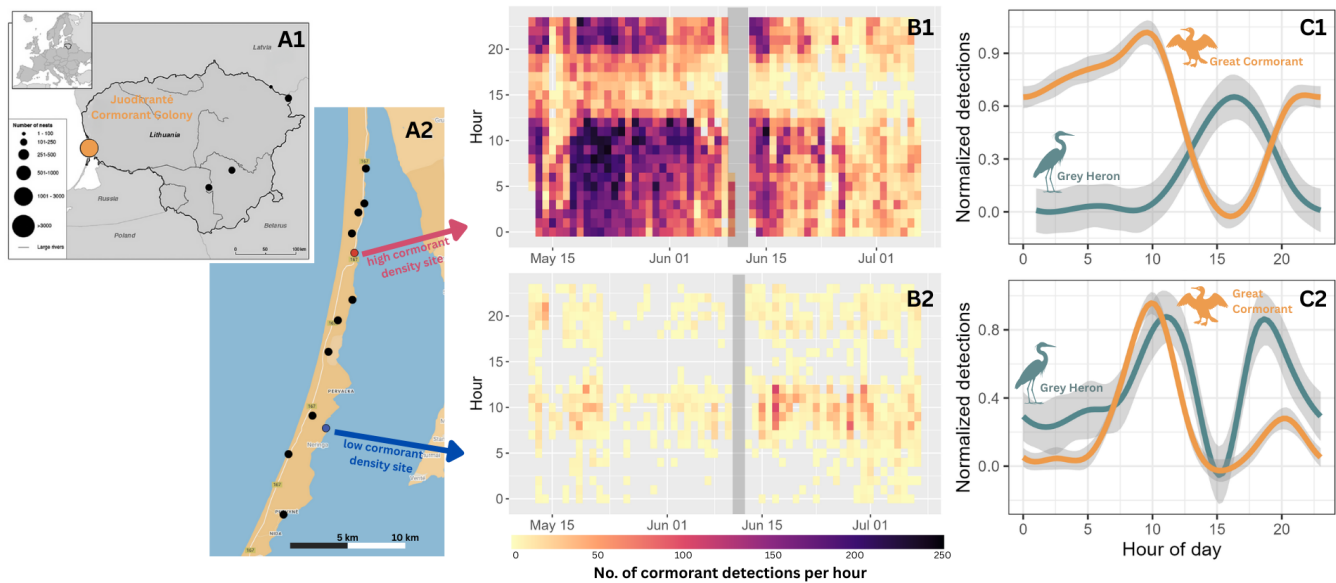
Changes in the ecosystem can affect the behaviour of birds and mammals co-existing within cormorant colonies. Fish regurgitated by cormorants, as well as juvenile cormorants falling from nests, provide an additional food source. Grey Herons (*Ardea cinerea*) may become commensals and exploit these food resources⁸. In addition, increased carrion availability attracts vertebrate predators and scavengers, resulting in higher densities of red foxes (*Vulpes vulpes*), raccoons (*Nyctereutes procyonoides*), and crows in the vicinity of cormorant colonies⁸. Similarly, some carnivorous birds, such as White-tailed Eagles (*Haliaeetus albicilla*) and White Storks (*Ciconia ciconia*), may disturb nesting cormorants and feed on regurgitated food⁷. Finally, standing defoliated trees created by cormorant nesting activity are used by woodpeckers as nesting site⁸.

Although the impacts of Great Cormorants on co-existing bird and mammal communities are increasingly recognized, how these impacts alter fundamental behavioural patterns, such as diel and seasonal activity, remains understudied. Some species may shift seasonal activity according to breeding seasons⁶, while others may shift their diel activity to avoid or exploit cormorant presence. However, investigating such changes in activity requires fine-scale, broad-coverage, cross-taxonomic monitoring. Recent advances in passive monitoring technologies, including autonomous recording units (ARUs) and camera traps (CTs), have overcome many of these limitations by enabling continuous, low-cost data collection across large spatial and temporal scales^{9–13}. In parallel, developments in machine learning have greatly improved the efficiency of data processing, with tools such as BirdNET for acoustic data¹⁴ and MegaDetector for imagery¹⁵. Together, the integration of ARUs and CTs provides an unprecedented opportunity to examine cross-taxonomic diel and seasonal activity patterns and thereby offering new insights into the ecological consequences of cormorant colonies.

In this research, I will investigate the diel and seasonal activity of bird and mammal communities inside and around the Juodkrantė cormorant colony, the largest colony in Lithuania, consisting of more than 4,000 breeding pairs and covering 14 ha within Curonian Spit National Park. Data will be collected ARUs and CTs, distributed across the cormorant density gradient. Specifically, the goals of this research are: 1) To quantify differences in the density of avian species, such as Grey Herons and woodpeckers across a gradient of cormorant density, and to examine shifts in their diel and seasonal activity in relation to cormorant activity; 2) to quantify the relationship between cormorant colony intensity and shifts in the diel and seasonal activity patterns of local mammal communities, including red foxes, raccoon dogs (*Nyctereutes procyonoides*), moose (*Alces alces*), wild boar (*Sus scrofa*), and roe deer (*Capreolus capreolus*)⁶; and 3) to provide a general workflow for integrating ARUs and CTs for cross-taxonomic monitoring and the study of species interactions. Ultimately, this research will improve our understanding of how Great Cormorants shape the behavior and interactions of co-occurring wildlife and provide a scalable framework for cross-taxonomic monitoring of species interactions in complex ecosystems.

Preliminary results: A pilot field season was conducted with local Lithuanian partners from May to July 2025, during which ARU sites were established across Curonian Spit National Park, including the main Great Cormorant colony (Fig.A1 & A2). Preliminary results suggest that Grey Herons at high cormorant density sites (Fig.B1) exhibit a unimodal activity pattern, with peak activity in the afternoon, coinciding with the cormorants' feeding period (Fig.C1), whereas herons at low-density sites (Fig.B2) show a bimodal pattern, with peaks in the late morning and evening (Fig.C2).

Funding status: The 2025 field season was funded by National Geographic Early Career Grant (EC-KOR-53792R-18). The 2026 field season, including equipment support, will be supported by a Wildlife Acoustics Grant; while additional funding, Baltic Summer University Scholarship, is currently under application. Postdoctoral research funds will be partially used to cover fieldwork expenses not supported by other grants.



▲ **(A1)** Locations of main Great Cormorant colonies in Lithuania, highlighting the largest colony, Juodkrantė (orange dot), within Curonian Spit National Park. **((A2))** Curonian National Park, with black dots indicating pilot study sites set up in 2025. **(B1–B2)** Hourly and daily vocal activity patterns of Great Cormorants at high-density **(B1)**, near the colony core) and low-density **(B2)**, further away from the core) sites. **(C1–C2)** Diel activity patterns of Great Cormorants and Grey Herons at high-density **(C1)** and low-density **(C2)** sites.